

WARRNAMBOOL, Australia

WMO: 948370

Lat: 38.2867S

Lon: 142.4522E

Elev: 234

StdP: 14.57

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.9	38.2	34.0	28.9	40.7	35.7	31.0	42.1	30.4	50.9	28.6	50.8	6.8	10	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.0	90.9	66.8	85.1	65.5	79.8	64.3	69.5	82.4	67.4	79.8	65.6	76.9	15.5	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.6	95.4	72.5	63.5	88.5	70.3	61.8	83.3	68.9	33.7	82.1	32.0	80.0	30.6	77.2	91.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.9	26.0	23.0	31.7	102.5	1.6	4.8	30.6	106.0	29.6	108.8	28.7	111.5	27.5	115.0
			31.4	73.7	1.6	2.8	30.2	75.8	29.3	77.4	28.4	79.0	27.3	81.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.8	63.0	64.1	61.5	57.4	53.3	49.7	48.8	49.7	51.5	54.1	57.2	60.0	(6)
	DBStd	7.29	6.73	6.51	5.98	5.02	3.97	3.21	2.93	3.47	3.89	5.26	6.04	6.40	(7)
	HDD50	195	0	0	0	1	11	43	57	48	22	11	2	0	(8)
	HDD65	3581	120	88	148	238	363	458	501	476	404	343	252	190	(9)
	CDD50	2313	404	394	356	222	113	34	21	38	68	137	216	310	(10)
	CDD65	226	59	62	39	9	1	0	0	0	0	4	17	35	(11)
	CDH74	2632	708	616	416	109	5	0	0	0	1	80	245	452	(12)
	CDH80	1120	344	277	168	29	0	0	0	0	0	18	77	207	(13)
(14) Wind	WSAvg	11.5	12.0	11.7	11.0	10.2	11.1	10.7	12.2	12.2	11.9	11.9	11.6	11.9	(14)
(15) Precipitation	PrecAvg	30.5	1.3	1.2	1.9	2.4	2.9	3.2	3.7	3.8	3.2	2.8	2.4	2.0	(15)
	PrecMax	40.0	4.4	3.6	6.0	5.5	8.0	6.6	6.4	8.0	6.3	5.4	4.4	5.6	(16)
	PrecMin	19.2	0.2	0.0	0.4	0.3	0.6	0.7	1.6	0.7	1.3	0.6	0.6	0.1	(17)
	PrecStd	5.1	0.8	0.9	1.1	1.4	1.5	1.3	1.2	1.4	1.2	1.3	1.1	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.0	96.2	92.6	84.1	71.8	62.7	60.5	66.2	72.4	82.3	87.8	95.7	(19)
		MCWB	68.7	67.9	65.4	61.7	58.5	56.5	54.0	57.3	60.4	65.0	66.3	66.5	(20)
	2%	DB	89.3	89.0	84.3	76.5	66.2	59.2	57.7	61.7	66.7	74.8	81.1	85.5	(21)
		MCWB	67.0	67.7	64.8	60.2	57.1	54.6	52.8	54.4	57.7	62.6	64.8	65.1	(22)
	5%	DB	81.7	81.6	77.8	70.8	62.9	57.3	56.0	58.6	62.7	69.5	75.0	77.6	(23)
		MCWB	65.3	65.9	63.6	59.2	56.4	53.2	51.7	52.9	55.7	60.0	63.3	63.5	(24)
	10%	DB	74.8	75.4	71.9	66.1	60.2	55.6	54.4	56.2	59.5	64.3	68.8	71.8	(25)
		MCWB	63.9	64.3	61.7	58.1	55.3	52.2	50.7	51.6	54.0	57.4	60.9	62.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.4	72.9	69.5	64.3	61.0	58.2	55.7	58.2	61.4	67.2	69.6	70.0	(27)
		MCDB	86.6	84.3	80.9	77.3	65.7	60.3	58.5	65.0	69.1	77.4	81.9	84.4	(28)
	2%	WB	69.1	69.0	66.4	62.1	58.9	55.5	54.0	55.3	58.5	63.3	66.4	66.8	(29)
		MCDB	82.7	81.7	79.1	70.8	63.6	58.1	56.4	60.0	65.1	73.1	76.8	78.8	(30)
	5%	WB	66.6	66.8	64.5	60.5	57.4	53.9	52.6	53.6	56.4	60.5	63.6	64.6	(31)
		MCDB	79.1	79.1	74.9	68.1	61.6	56.4	55.0	57.5	61.7	68.8	73.2	75.8	(32)
	10%	WB	64.4	64.8	62.8	58.9	55.8	52.6	51.3	52.2	54.5	57.7	61.0	62.4	(33)
		MCDB	74.1	74.1	70.5	65.1	59.5	55.0	53.7	55.4	58.8	63.4	68.9	72.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	19.3	18.0	17.5	16.7	13.4	12.2	11.6	12.8	14.7	17.1	17.6	18.9	(35)
		MCDBR	33.3	30.1	29.3	26.3	18.5	14.6	13.5	17.1	21.0	28.0	29.9	32.3	(36)
		MCWBR	13.3	11.9	12.5	12.6	11.6	10.7	9.7	11.3	13.4	16.4	15.5	14.3	(37)
	5% WB	MCDBR	27.3	24.6	24.0	21.3	15.6	12.6	12.1	15.1	18.8	26.1	26.9	27.4	(38)
		MCWBR	12.9	11.9	11.5	11.9	10.8	10.0	9.5	10.8	12.7	16.3	15.3	13.6	(39)
(40) Clear-Sky Solar Irradiance	taub		0.346	0.340	0.329	0.317	0.309	0.302	0.299	0.312	0.336	0.341	0.342	0.337	(40)
	taud		2.494	2.529	2.560	2.578	2.577	2.581	2.586	2.541	2.454	2.448	2.475	2.512	(41)
	Ebn at Noon		313	307	298	282	264	257	265	279	290	303	312	317	(42)
	Edh at Noon		36	33	30	25	22	20	22	26	32	36	36	36	(43)
(44) All-Sky Solar Radiation	RadAvg		2263	1945	1531	1065	708	563	620	889	1275	1707	1993	2241	(44)
	RadStd		128	106	76	71	34	42	38	47	79	121	118	137	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.70	N/A	N/A	+1.91	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (26 neighbors)	+0.81	N/A	N/A	+1.57	+0.70	N/A	N/A	-211	+270	+72	(47)

Nomenclature: See separate page